

DNA Sequence Analysis Project

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Objective

This notebook demonstrates a reproducible analysis of DNA sequences by calculating the GC Content using BioPython.

```
In [ ]: from Bio.Seq import Seq

# Define DNA sequence
dna = Seq("ATGGCGCGCTTA")

# Calculate GC Content percentage
gc_content = (dna.count("G") + dna.count("C")) / len(dna) * 100

# Display Results
print("DNA Sequence:", dna)
print(f"GC Content: {round(gc_content, 2)}%")
```

Conclusion

The GC content of the target sequence was successfully calculated. The sequence exhibits a GC ratio of 58.33%, which reflects its thermal stability profile in DNA strands.